

7(a)	any four from: 1 respiration rate decreases / less ATP produced ; 2 reduces activity of (named) enzymes / description ; 3 reduces movement of, Ca^{2+} / Na^{+} / ACh / neurotransmitter ; 4 (<i>in neuromuscular junction</i>) less / no, acetylcholine broken down acetylcholine remains attached to receptors acetylcholinesterase, less active / inactive ; 5 (<i>in sarcomere</i>) fewer / no, Ca^{2+} bind to troponin fewer / no, cross bridges formed fewer / no, power strokes ; 6 (<i>in sarcomere</i>) less / no, detachment of myosin heads (from actin) fewer / no, cross bridges broken ATPase / myosin head, less active / inactive hydrolysis of ATP, slower / stops ; 7 AVP ; e.g. shivering / reduced blood flow to muscle / energy diverted for thermoregulation (<i>idea</i>)	4
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Question	Answer	Marks
7(b)	any three from: 1 Na⁺ channels don't open A Na⁺ do not, diffuse / move, into neurone or stay open ; A Na⁺ always, diffuse / move, into neurone 2 (axon) <u>membrane</u>, cannot be depolarised or (axon) <u>membrane</u> stays depolarised ; 3 no action potentials, generated / transmitted (along motor neurone) or continuous action potentials, generated / transmitted (along motor neurone) ; 4 Ca²⁺ does not enter presynaptic knob / vesicles not stimulated to move / no ACh release (at neuromuscular junction) or Ca²⁺ always enters presynaptic knob / vesicles always stimulated to move / ACh continually released (at neuromuscular junction) ; 5 consequence e.g. paralysis / described or continued muscle contraction ;	3